2-step PCR mixture and conditions (Barcoded-head primers for seqs pooling)

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Abstract

PCR mixture and condition (2X SUPERGREEN PCR MASTER MIX)

Procedure

Step 1

Wear glove, clean up the working bench w. 1% bleach

For 1' PCR head-primers

Step 2

Prepare 1' PCR master mixutre for **head-primers** (**prepare 1.2X of solutions for pipetting error if needed**)

PCR mixture for head-primers for each reaction

Component	Volume	Volume (1.2X)	Final conc.
PowerPol 2X PCR Master Mix	12.5 µL	15 µL	1X
Forward Primer (10 µM)	0.5 µL	1.2 µL	0.2 µM
Reverse Primer (10 µM)	0.5 µL	1.2 µL	0.2 µM
ddH2O	10.5 µL	11.1 µL	-
Total volume	24 µL	28.5 µL	

Component	Volume	Volume (1.2X)	Final conc.
2X KAPA HiFi HotStart ReadyMix	12.5 µL	15 µL	1X
Forward Primer (10 µM)	0.75 µL	0.9 µL	0.3 µM
Forward Primer (10 µM)	0.75 µL	0.9 µL	0.3 µM
ddH2O	10 µL	12 µL	-
Total volume	24 µL	28.8 µL	-

Note

Negative control ALWAYS NEEDED! For example, if you have 5 PCR reactions to run, prepare master mixture for 6 reactions (5 DNA template + 1 negative control).

Step 3

Mix the 1' PCR master mixture gently by pippeting. Quick spin the tube.

Step 4

Transfer **24 µL** 1’ PCR master mixutre in 8-strip PCR tubes.

Step 5

Add **1 µL** DNA template in 8-strip PCR tubes, resulting in a **25 µL** reaction mixture for 1’ PCR.

> **Note**
>
> **Negative control** contains only **24 µL** master mixture but not DNA template

Step 6

Mix the reaction mixture gently by tapping the tubes. Quick spin the tubes.

Step 7

Carry out PCR using the following condition:

1’ PCR condition for **head-primers**

Step	Temp	Sec	Cycle
Initial denaturation	95 ºC	180	
Denaturation	95 ºC	30	25 cycles
Annealing	60-66 ºC varied (b)	30	
Extension	72 ºC	180	
Final extension	72 ºC	420	
Preservation	Preservation	4 ºC	∞

b. Annealing varied, 60-66C is working; Refer to 1’ PCR primers for annealing temperature c. 1kb ~ 1min extension; enough time allow full extension of sequence

Step 8

Carry out **electrophoresis** for inspection of DNA products

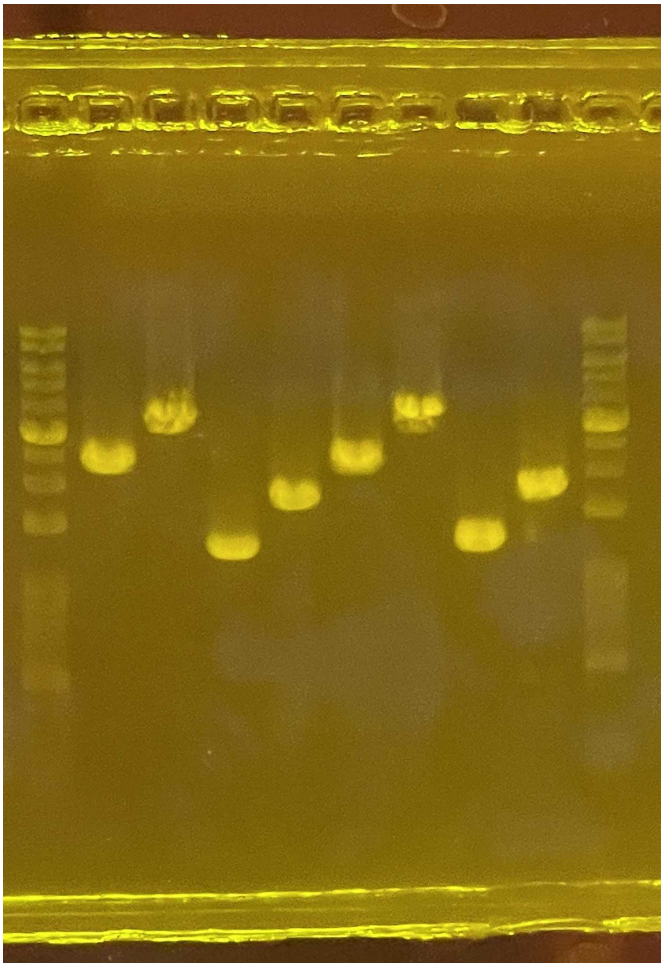


Figure 1: Gel before markdown

Step 9

Markdown wells and upload the pictures to the Lab Google drive

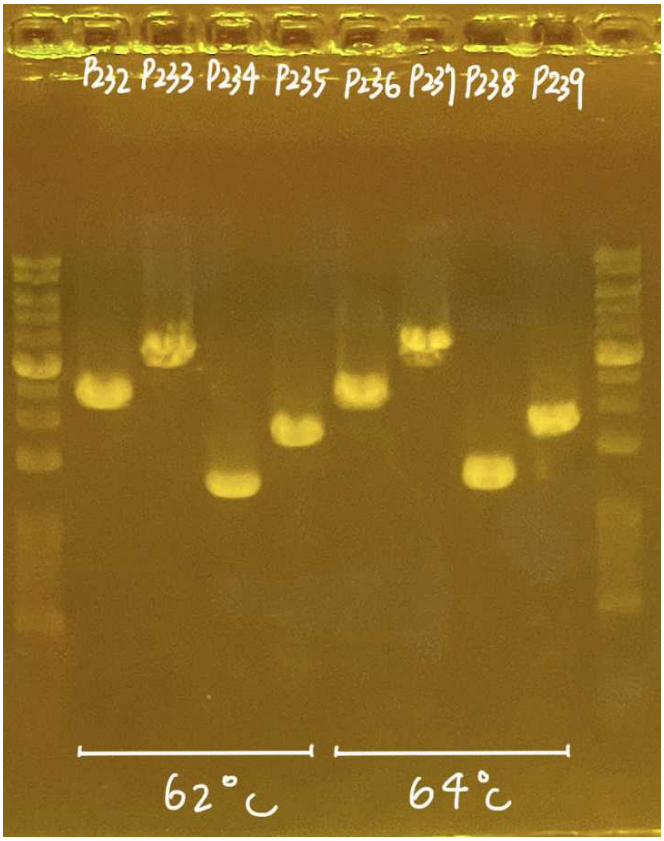


Figure 2: Marked gel picture go to the Lab Google drive

For 2’ PCR barcoded-head primers

Step 10

Prepare 2’ PCR master mixutre for **barcoded-primers** (**prepare 1.2X of solutions for pipetting error if needed**)

PCR mixture for barcoded-primers for each reaction **(NO PRIMERs at this point!!)**

Component	Volume	Volume (1.2X)	Final conc.
ZEJU PCR Master Mix	7.5 µL	9 µL	-
ddH20	5.55 µL	6.66 µL	-
Total volume	13.05 µL	15.66 µL	-

Note

Negative control ALWAYS NEEDED! For example, if you have 5 PCR reactions to run, prepare master mixture for 6 reactions (5 DNA template + 1 negative control).

Step 11

Mix the 2’ PCR master mixture gently by pippeting. Quick spin the tube.

Step 12

Transfer **13.05 µL** of the 2’ PCR master mixture to 8-strip PCR tubes.

Step 13

Add **1.2 µL pre-mixed barcoded-head primers** (Forward + Reverse) to each PCR tubes.

Step 14

Add **0.75 µL** of **1’ PCR product as template**, resulting in **15 µL** reaction mixture for 2’ PCR.

Negative control contains only **14.25 µL** master mixture and premixed barcoded-head primers but not DNA template

Step 15

Mix gently by tapping the tubes. Quick spin the tubes.

Step 16

Carry out 2’ PCR using the following condition:

2’ PCR condition for **barcoded-head primers**

Step	Temp	Sec	Cycle
Initial denaturation	98 ºC	30	
Denaturation	98 ºC	15	12 cycles
Annealing	64-68 ºC varied (a)	15	
Extension	72 ºC	20 (b)	
Final extension	72 ºC	210	
Preservation	Preservation	4 ºC	∞

a. Annealing varied, 65 C is working based on test on 220531; Refer 2’ PCR primers for annealing temperature b. 1kb ~ 1min extension; enough time allow full extension of sequence

Step 17

Carry out **electrophoresis** for inspection of DNA products

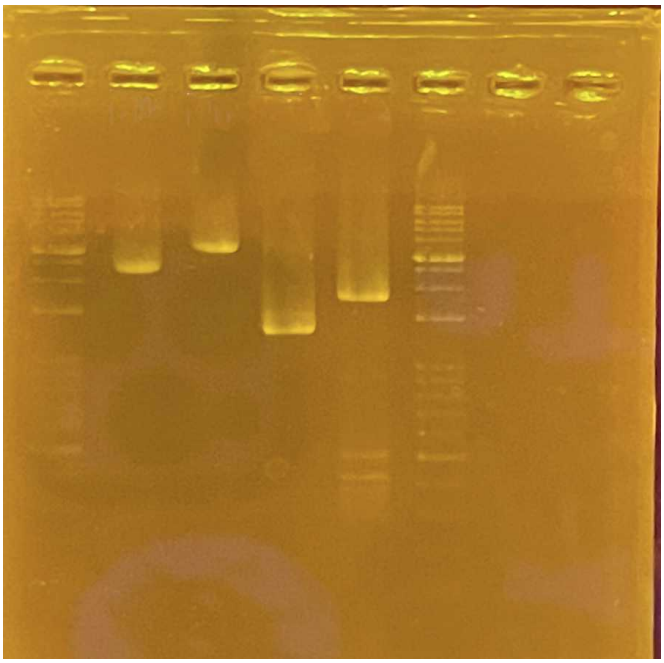


Figure 3: Gel before markdown

Step 18

Markdown wells and upload the pictures to the Lab Google drive

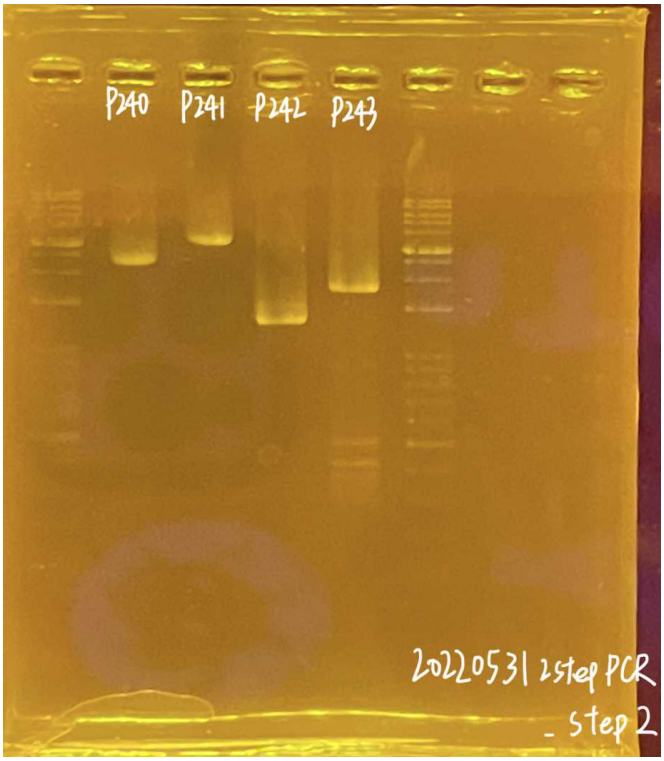


Figure 4: Marked gel picture go to the Lab Google drive